

Deep Cycle Solar Energy Lithium Ion Battery for Solar Storage System and Telecom

User Manual

4850

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1. Brief Introduction

Li-ion battery system in solar storage system or Telecom is a high-tech product which is researched and developed by SACRED SUN. With its integration, miniaturization, light-weight, intelligent centralized monitoring, battery maintenance and management, unattended, convenient use of the standardization of cabinet installation, energy conservation and environmental protection, are widely applied in remote access network equipment, remote switch unit, mobile communication, transmission equipment, satellite ground stations and microwave communication equipment and other areas of the communication as a backup power supply. Currently, the battery system is in bulk production status in domestic which are widely used as backup power in the field of solar storage system or telecomm, and also got favor in the solar storage system or telecomm industry.

2. Product Feature

- (1) Battery adopts LiFePO₄ as cathode material;
- (2) The system adopts high performance of BMS battery management module, The BMS has the protection function in current, voltage and temperature; it also makes a good communication between battery system and the host equipment;
- (3) Monitoring unit will automatically measure the battery charge and discharge current, charge and discharge voltage, single cell surface temperature and environmental temperature;
- (4) Power cut off function. It will have the alarm information if the battery voltage is lower than the alarm value, when the voltage is too low, it will protect the batteries by automatic power cut off;
- (5) The system has a good electromagnetic compatibility;
- (6) With intelligent design, equipped with centralized monitoring module, four control functions (remote sensing, remote communication, remote control and remote adjustment), it realizes the computer management, can communicate with the central remote control center;
- (7) Effective combinations of power control technology with computer which realizes real-time monitoring and control all kinds of parameters and state;
- (8) Using self cooling method and the whole system is with an extremely low level noise;

3. Product advantage

- (1) It can be directly implemented in the original communication DC switching power supply system of charging and discharging with the same working pattern;
- (2) Battery system in solar storage system or telecom is a high-tech product which successfully designed and developed by SACRED SUN, with independent intellectual property rights, the product fill the gap of domestic battery industry, the main technology is kept in the international leading level;
- (3) Communication battery system is in bulk production in domestic, and is widely used in the solar storage system or telecom as back-up power system, it has long and mature market application, and won recognitions from market.
- (4) To meet solar storage system or telecom operator's new requirements of the distributed power system, SACRED SUN proposes integrated back-up power solutions based on the LiFePO₄ batteries, the BMS and DC switching power supply module (or UPS uninterruptible power supply system).
- (5) High volume energy density weight energy ratio, there is no special requirements for space and weight so as to reduce the rent cost.
- (6) LiFePO₄ cell life cycle could reach 3500 time RT 80% DOD
- (7) High working ambient temperature, working temperature could be in -20~+60°C, recommended working temperature is -20~+50°C, greatly reduce the cost of equipment.

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(8) LiFePO₄ single cell adopts fully sealed structure, can be placed in random, no harmful gases and no pollution, and it is environmental protection product.

(9) LiFePO₄ battery has high discharge rate performance; even a smaller capacity of LiFePO₄ battery system can meet the application environment of large current discharge.

(10) Flexible configuration: multiple battery modules could be used in parallel to make communication battery system of large capacity into practical application, both is to increase the output power of the system, and extend the backup time of the system.

(11) Intelligent display: configured intelligent display system, can real time to check the battery status and data, it will automatic become black screen when not in use the state, keeping the maximum capacity.

4. Product specifications and technical parameters

Table 4-1 48V series product specifications

Model	Voltage (V)	capacity (Ah)	maximum discharge current (A)	Dimension (mm) (w×h×L)	weight (kg)	note
4850	48	50	50	442×132×440 (3U)	32.0	module
48100	48	100	100	442×222×395 (5U)	62.0	module

Notes:

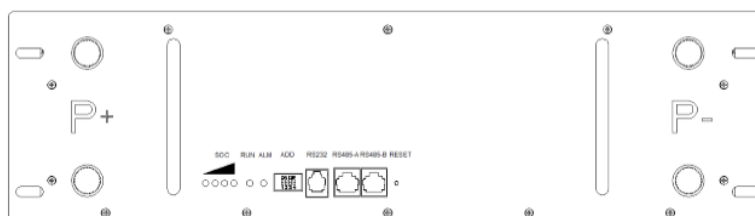
- Battery supports to use in parallel, but only could extend the time of backup after parallel; We suggest that the maximum current of load should not increase with an increase of the number in parallel.
- The upper table is only for the size of the battery system, not include the size of cabinets, racks, rain box.

5. Environmental Requirements

- environment temperature: -20~+60°C (Recommend Operating Temperature: (-15°C~+50°C);
- Storage Temperature:-40°C~80°C
- Relative humidity: 5%~95%RH;
- Not over 3000 meters above altitude
- No conductive dust and corrosive gas around.

6. Panel function of system

6.1 3U height of crate system schematic



6-1 Traditional system panel diagram

6.2 Battery output terminal

“1”stands for output terminal; Series of specifications for 48 v battery system products adopt 4 core terminal blocks, mainly consider when the battery module is used in cascade, the 4 pins in the connection terminals from left to right is defined as the battery anode ,cathode, anode and cathode.

Table 6-1 Different spec batteries, output port connection configuration instructions

Battery model	Battery capacity	Terminal specifications	Diameter of wire	Connecting lug
4850	50AH	M10	≥16mm ²	25-10
48100	100AH	--	≥25mm ²	25-10
100AH above	--	--	≥25mm ²	25-10

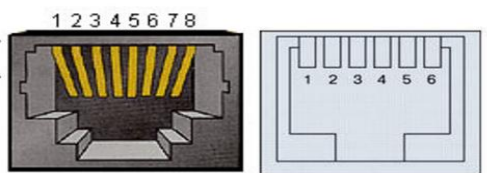
6.3 Reset Key ON/OFF Key

“RESET “Key. When system is abnormal, this key can reset the system to ensure the stability of the running system. After BMS is reset, the data and function set from upper computer is still kept, if need to recover to original value, can be set on the upper computer, but there is no change in running record and storage data, such as capacity, cycle life and protection record state. Details of function are defined in the following table.

Table 6-2 Definition ON/OFF Key

#	Function	Definition	Note
1	ON/OFF	When BMS in dormant status, press the button for 3s then the BMS will be activated, the running light will be on for 0.5s	
		When BMS in active status, press the button for 3s then the BMS will be in dormant status, the capacity light will be on from the lowest capacity for 0.5s	
		When BMS in active status, press the button for 6s then the BMS will be reset, all LED light will be on for 1.5s	

6.4 Communication Connection



RS485 Interface

RS 232 Interface

6.4.1 First class cascading communication connection

“3”is RS485 communication port. when the system is cascade, data transmission is through RS485 serial communication port. Table 6-3 is definition of RS485 Port.

Cascade pack internal system use RS485 communication port, upper computer will collect each data of Slave Pack through Master Pack.

Table 6-3 RS485 definition of Port

Lead foot	Description
1、 8	RS485-B
2、 7	RS485-A
3、 6	Land
4、 5	NC (hang in the air)

6.4.2 On Position Communication Connection

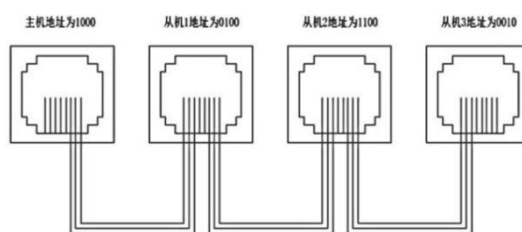
“4”is RS232 superior communication connection. Battery system use RS232 serial port communication, definition of RS232 communication port is shown in table 6-4.Through RS232 serial port, can centralize monitor the background of battery system ,it will transfer the information such as the battery system working status and alarm information to the remote monitoring center, realize remote monitoring. When using RS232 communication, baud rate should be set to 9600 BPS, set the dial code switch of the battery system to Pack1 (Master Pack), then RS232 communication module is valid and can communicate with the upper computer.

Table 6-4 RS232 definition of port

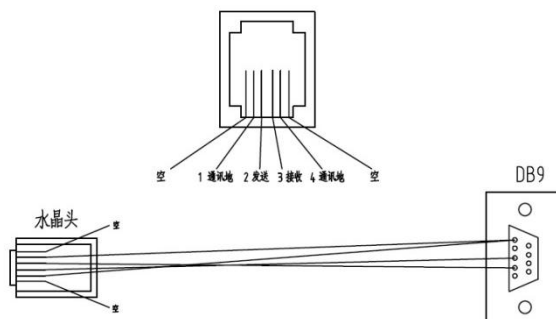
RJ11 Lead foot	Description	DB9 Pin foot	Description
3	BMS TX	2	RXD
4	BMS RX	3	TXD
5	land	5	land
1\2\6	NC	other	NC

6.4.3 Communication wires

Graph 6-2 is cascade wiring diagram; Graph 6-3 is RS232 upload data communications illustration chart in which there is no connection in the head first of the six-core crystal head and sixth welding pieces.



6-2 RS485 cascade communication diagram



6-3 RS232 communication cable

6.5 Dial Switch

“5”stands for dial switch, there are four dial code switches on panel which are used to set the address code in the modules in communication, in accordance with the system address from 0 to 15, dial code switch from left to right allocation as below:



6-5 Address of dial switch

Address #	Dial switch Position				Remark
	#1	#2	#3	#4	
0	OFF	OFF	OFF	OFF	no cascade, one machine use
1	ON	OFF	OFF	OFF	set as Pack1 (main)
2	OFF	ON	OFF	OFF	set as Pack2
3	ON	ON	OFF	OFF	set as Pack3
4	OFF	OFF	ON	OFF	set as Pack4
5	ON	OFF	ON	OFF	set as Pack5
6	OFF	ON	ON	OFF	set as Pack6
7	ON	ON	ON	OFF	set as Pack7
8	OFF	OFF	OFF	ON	set as Pack8
9	ON	OFF	OFF	ON	set as Pack9
10	OFF	ON	OFF	ON	set asPack10
11	ON	ON	OFF	ON	set asPack11
12	OFF	OFF	ON	ON	set asPack12
13	ON	OFF	ON	ON	set as Pack13
14	OFF	ON	ON	ON	set as Pack14
15	ON	ON	ON	ON	set as Pack15

The factory default dial code switch position is address 0, which is 0000.

6.6 LED Indicator description

(1) The capacity light (SOC, Green)

The capacity light: 4 green lights, each indicator light stands for 25% of capacity, capacity of 100%, the four lights all light up; Capacity is 75%, the first on the left, the lights went out on and right side of the three lights light on; Capacity is 50%, the left two lights went out and right side of the two lights light on; Capacity is 25%, the left three lights went out and a light on the right light on.

Status		Charge				Discharge			
capacity light		L4	L3	L2	L1	L4	L3	L2	L1
Power (%)	0~25%	off	off	off	flash 2	off	off	off	on
	25~50%	off	off	flash 2	on	off	off	on	on

	50~75%	off	flash 2	on	on	off	on	on	on
	75~100%	flash 2	on	on	on	on	on	on	on
running light 		on				flash (flash 3)			

Flashing way:

Flashing way	on	off
Flash 1	0.25S	3.75S
Flash 2	0.5S	0.5S
Flash 3	0.5S	1.5S

(2) Running light (RUN, Green)

RUN light, Green, will be light on all the time, when charging 0.5s light on and 0.5s light off when discharging.

(3) The alarm indicator light (ALM, Red)

ALM light, red, it will light on if a fault was occurred

Definition of specific indicator light, shown in the table below:

Table 6-6 LED indicator light illustration ( means light on o mean light off)

status	normal/warning/protection	RUN	ALM	capacity LED				remark
								
power off	dormant	off	off	off	off	off	off	all are off
stand by	normal	flash 1	off	shown according to capacity				standby status
	warming	flash 1	flash 3					module low pressure
charge	normal	on	off	shown according to capacity				lightest capacity LED flash （flash 2） , overcharge warning ALM not flash
	warming	on	flash 3	(lightest capacity LED flash2)				
	overcharge protection	on	off	on	on	on	on	if no electricity, light will change to be standby status
	temp、over current、fault protect	off	on	off	off	off	off	stop charging
discharge	normal	flash 3	off	shown according to capacity				
	warming	flash 3	flash 3					
	under voltage	off	off	off	off	off	off	stop discharging

	protection							
	temp、 over current、 short circuit、 converse protection fault protect	off	on	off	off	off	off	stop discharging
out off operation		off	on	off	off	off	off	stop discharge and charge

7、 System working principle and parameter

7.1 System working principle

Working principle of 4850 backup power system in solar storage system or telecom: input electricity power 220V, after processing power rectifier module, there is - 48V output, under the normal power grid, power rectifier module provides power to system, at the same time to charge the battery system; when in power grid cut off condition, battery system will provide electrical power to ensure the normal operation of the DC power supply system so as to realize the uninterrupted power supply function. Power supply has second time power down function, if the battery voltage is too low, the power will cut off the supply of battery power so as to protect the battery life.

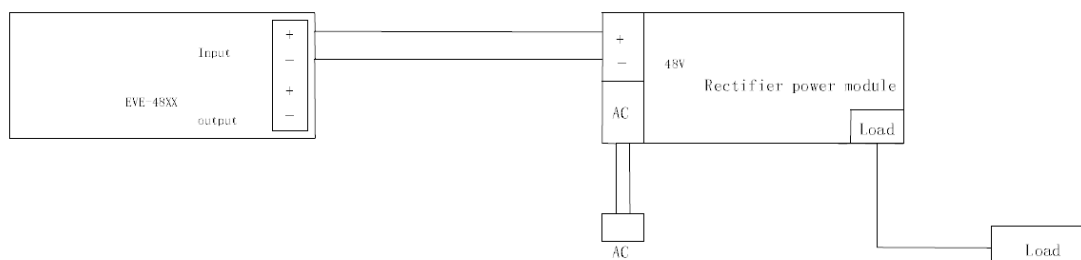


Table 7-1 Li-ion Battery system in solar storage system or telecom working diagram

7.2 Charging parameter

Table 7-1 Charging Parameter

Spec, model	Charging voltage (DCV)			Charge current limit (A)
	Min value	Typical value	Max value	
EVE-48XX Series(15S)	53.2	54	54.8	Charge current limit, typical 0.2C ₃

7.3 Discharge Parameter

Table7-2 Partial product Partial discharge parameters

Spec, model	Discharge voltage(DCV)			Discharge current(A)	
	Min value	Typical value	Max value	Typical value	Max value
48XX series(15S)	40.5	48	54	0.2C	1C

7.4 System protection function and parameters

Table 7-3 System protection function and parameters

NO	Protect Item		Default
			15S
1	system total high voltage protection and recovery	Alarm value	54.00V
		Protection value	54.8V
		protection Recovery value	51.7V
2	system total low voltage protection and recovery	Alarm value	43.5V
		Protection value	40.5V
		protection Recovery value	43.5
3	short circuit protection	Protection value	≥180A
4	unit cell low voltage protection and recovery	Alarm value	2.9V
		Protection value	2.7V
		protection Recovery value	2.9V
5	unit cell high voltage protection and recovery	Alarm value	3.55V
		Protection value	3.70V
		protection Recovery value	3.45V
6	Charging over current protection (No charge current limit)	Alarm value	50A
		Protection value	55A
		protection Recovery value	1min recover or discharge recover
7	Discharge over current protection	Alarm value	-55A
		Protection value	-60A
		protection Recovery value	Charge recover
8	Cell high temperature protection and recovery	Charging high temperature protection value	70°C
		Charging high temperature recovery value	60°C

		discharging high temperature protection value	70°C
		discharging high temperature recovery value	50°C
9	Cell low temperature protection and recovery	Charging low temperature protection value	-5°C
		Charging low temperature recovery value	0°C
		discharging low temperature recovery value	-20°C
		discharging low temperature recovery value	0°C
10	Environmental temperature protection and recovery	high ambient temperature alarm value	65°C
		low ambient temperature alarm value	-20°C
		high ambient temperature protection value	70°C
		high ambient temperature recovery value	65°C
		low ambient temperature protection value	-25°C
		low ambient temperature recovery value	-20°C

Various protection functions and parameters of battery system

Remark: The above table is the default parameters, set of the specific parameters may be different due to some models may be configured according to the actual needs of customers please refer to the actual product parameters.

8. System installation, use and maintenance

8.1 System installation

8.1.1 3U Height system installation instruction



Table 8-1 5U system installation structure diagram

(1) Recommend to install in the 3U standard cabinet, can also be installed on the wall, to fix on the machine frame with four M6 bolts on both sides of installation ear of the cabinet.

(2) connect the grounding hole of the Ear hanging on the cabinet panel with 2.5mm yellow green cable for grounding, grounding to ensure good.

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(3) capacity in or below 20Ah battery system ,connect the switch power supply or the device of positive and negative pole to the battery positive and negative electrode of output terminals with more than 2.5 mm² red and black cable respectively. While capacity above 20Ah, connect the switch power supply or the device of positive and negative pole to the battery positive and negative electrode of output terminals with more than 2.5 mm² cable respectively, pay attention to the positive and negative signs.

8.1.2 Bracket mounting

3U height is the dimensions of 50Ah Li-ion battery in solar storage system or telecom system, can be respectively connected in parallel to form all kinds of 48V series product specifications, to connect the support bus copper bar on the "+" "-" terminal with each 3U battery output terminals "+" "-" pole in the cabinet respectively using above 10.0mm long cable black and red two cables to form above 100Ah battery system.

8.1.3 Cabinet type installation

Cabinet installation, it is for requiring high dustproof indoor cell system's application scenarios, generally by the 19 inch standard cabinet, battery pack, bus bar, cable and other components, 19 inch cabinet height can be configured according to the number of cells, the installation sequence is as follows:

- (1) With the above 10.0mm long, two cables separately connect the battery system "+" - "to the parallel bus copper bar on the top of the cabinet, red cable connect "+", black cable " - ";
- (2) Each battery module is fixed on the cabinet tray, and the hanging ear of each battery module is fixed on the bracket in the cabinet with M6 screw;
- (3) The grounding hole on the right side of the battery use more than 2.5 mm 2 pieces yellow green cable for grounding, to insure good grounding
- (4) The total positive and negative lines drawn out from the bus bar of the cabinet connect the two ends of the positive and negative poles with the switch power supply or the equipment.

8.2 System power up

After the installation of the system, it is generally in standby state; if needed, as long as the simple power on operation, it will be able to put the equipment into the normal operation.

8.2.1 System activation

In storage, transportation, or in the absence of external power supply and no load after 24 hours, the battery system has been in a state of sleep. Adding allowing the charging voltage, the battery system is activated into the normal operation state, it can be charged, discharged, or enters to the standby state. When the battery system is in a sleep state, the switching DC power, matching use with the battery system must cancel the anti reverse connection protection function, so as to activate the battery when recovering on AC with a 48V voltage to system.

8.2.2 System standby

After the system is activated, if the external power supply and the load are removed, the battery system will take the initiative to enter into standby state. In this state, the power consumption is very low, and the RUN lamp on the panel indicates that the battery system is in standby mode. 24 hours later, the battery system will automatically enter the sleep state. In this state, power loss will be lower, when the panel is all out of the indicating light; it indicates that the battery system is in a state of sleep.

8.3 Alarm instructions and processing

When the system is in failures, it will give an alarm signal; all the faults alarm is audible and visual and with text prompt. When alarm is on, the red warning light on the monitoring unit is on, and the alarm information is sent out to the remote monitoring center. When the system generates a warning, the remote monitoring center receive alarm information, maintenance personnel should immediately take corresponding measures according to the monitoring unit's alarm information, and do corresponding inspection equipment to determine the type and location of fault.

8.3.1 Alarm and Countermeasure of system output

In the system, if there is a single cell voltage which is over voltage, over current, under voltage protection, temperature protection and so on causing the normal output, please handle according to the table 8-1.

Chart 8-1 Alarm processing and Countermeasures

Status	Alarm item	Alarm indication	Handle method
Charging status	Cell over voltage	ALM light is on	Stop charging and look for the cause of the failure
	Charging Over current	ALM light is on	Stop charging and look for the cause of the failure
	Charging temperature protection	ALM light is on	Stop charging
Discharge status	Discharge over current protection	ALM light is on	Stop charging and look for the cause of the failure
	Discharge temperature protection	ALM light is on	Stop charging
	Total voltage under voltage protection	ALM light is on	charging
	Cell Voltage under voltage protection	ALM light is on	charging

8.3.2 Not affecting the system output alarm and countermeasures

When there is low voltage of the total voltage or the cell voltage is low, the battery system also generates the corresponding alarm signal. Maintenance personnel should check the equipment according to the prompt information, determine the fault type and location, and take appropriate measures to ensure that the system is in the best working state, so as to prevent the system output.

Its phenomenon and countermeasures are shown in table 8-2

Chart 8-2 Alarm but not protection

Alarm	FET action	alarm indication	countermeasure
Low capacity alarm	No	ALM Flashing	Stop discharge

8.4 Communication failure

In the PC software, you could see the status of the protection of the system so as to help analyze the specific causes of failure.

(1) When a communication failure occurs in RS232, you can follow the following steps to exclude the reason:

- Check communication line ;
- Reset Master Pack;
- For cascade systems, you can change the settings of Master Pack and Slave Pack, etc. to diagnose.

(2) When a communication failure occurs in RS485, you can follow the following steps to exclude the reason:

- Check communication line;
- Reset Slave Pack;

c. Exchange the setting of address of Slave Pack to diagnose

8.5 Protection of failure

8.5.1 over voltage protection

In the use of standard charging voltage, generally it will not appear cell inside is overpressure, unless there is a large imbalance between the cells. If the voltage is very uneven, there is cell over voltage, the total voltage is still relatively low, then there will be over voltage protection, you need to check the voltage of cell.

8.5.2 Battery protection

When the system's current is relatively small, and show current protection, you need to check the status instructions, check the connection line is normal or not.

8.5.3 Other failures

If you can't charge or discharge due to the protection alarm, you need to check the specific voltage, temperature and other parameters of the system and the status of the system to determine the cause of the failure.

8.6 Treatment of special circumstances

8.6.1 Power cut off

AC power is the most common in the system operation, when power cut is not long, DC power supply is provided by the system packed by Li-ion battery. If a blackout due to unknown reason or time is too long, need to pay attention to the power supply time of the system.

8.6.2 Catastrophic accident

Catastrophic accidents causing communication equipment failures are by lightning, flooding, earthquakes, fires and other disasters. A serious threat to the security of the communication of disaster, should take the possible prevention as priority, at the same time, communication station should cope with these disasters countermeasures and corresponding manpower and material force, and communication station should have a state of emergency management regulations and major accident repair rules.

9. Packaging, transportation and storage

9.1 Packing

System is with full packing which ensures that the products are not subject to any harmful gas, chemical contamination, electrostatic, wet and mechanical damage in handling, transportation and storage. The packing box has been marked with the name of the product, the type and size of the product, the date of production, the quantity and the batch number.

9.2 Transportation

Product is adapted to the car, boat, aircraft transport. When in transport, it should be covered to avoid the sunlight, and with civilized loading and unloading. With product packaging box allows using any kinds of transport, battery in loading and unloading process should be light moving gently to prevent throwing, rolling, pressing. Direct rain and mechanical impact of rain and snow should be avoided in transportation.

9.3 Storage

System storage should be kept in dry warehouse, not sun and rain. The harmful gases are not allowed in the warehouse, as well as flammable and explosive products and corrosive chemicals. TO avoid mechanical impact, stress and strong magnetic field effect, avoid direct sunlight and away from heat source not be less than 2m, the packing box should pad off the ground at least 20 cm high, away from the wall, window, or the air inlet at least 50cm. Under the provisions of the conditions of storage period of more than 3 months of products should charge once, storage period of more than 6 months products must check and test the capacity, store for more than 1 years of products must be re-examined, only can be used when is qualified.

10. Environmental protection requirements

To meet the RoHS certification requirements, cadmium free, hydride free, fluoride free or asbestos free, do not distribute organic compounds, packaging can be recycled.

11. Safety precautions

- (1) Prohibit connecting the battery into the water , prohibit to expose to the rain, should be placed in a cool dry environment when not in use;
- (2) Prohibit using or putting batteries near high temperature;
- (3) Make sure to use the battery system in accordance with the charging and discharging parameters specified in this specification;
- (4) Prohibit direct access to the power supply socket;
- (5) Battery is prohibited in the fire or heater;
- (6) Prohibit dismantling batteries and components;
- (7) Prohibit knocking, throwing or trampling battery etc.
- (8) Even if the city power cut, the system output interface is still with voltage output, user must pay attention to avoid electric shock or short circuit;
- (9) Please read the instruction manual carefully before use